

GNIOT Institute of Management Studies – PGDM (Batch 2025-27)**(TRIMESTER – I) END TERM EXAMINATION, OCTOBER-2025****Statistical Analysis for Managers****[Time: 1:00 Hours]****[Maximum Marks: 30]****Following are the course outcomes of the subject: - Statistical Analysis for Managers**

CO	Course Outcome (CO)
CO1	Understand the fundamentals of statistics and its interdisciplinary role in decision making
CO2	Present and understand the data
CO3	Perform fundamental univariate and bivariate data analysis to draw meaningful insights

*(Note- Faculty can add the more CO's as per their subject.)***SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, word limit 100-150 words)**

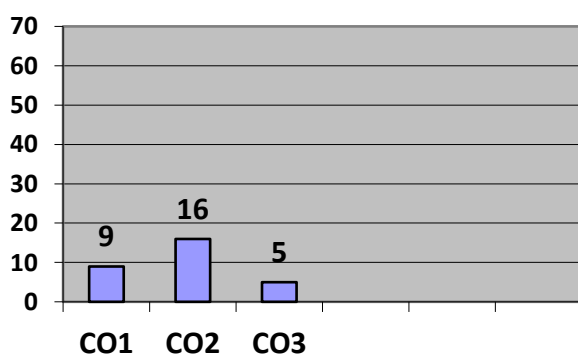
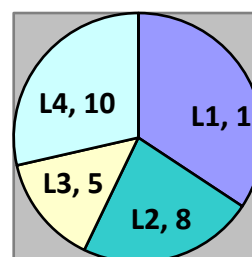
Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	3	CO - 1	BL1
(B)	3	CO - 1	BL1
(C)	3	CO - 1	BL1
(D)	3	CO - 2	BL1
(E)	3	CO - 2	BL2

SECTION-B**Attempt Any two questions out of the following: (Long Answer Type Questions, word limit 200-250 words)**

Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	5	CO – 2	BL3
2 (B)	5	CO –2	BL3
3 (A)	5	CO – 2	BL2
3 (B)	5	CO – 2	BL2

SECTION-C**Attempt any one parts of the following: (Caselet / Numerical, , word limit 200-250 words)**

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4(A)	5	CO – 3	BL 4
4(B)	5	CO –3	BL 4

Course Outcome Wise Marks Distribution**L1 L2 L3****Bloom's Level wise Marks Distribution**

Roll No:

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Subject Code: DPM104

Paper ID: 11004

GNIOT Institute of Management Studies – PGDM (Batch 2025-27)**(TRIMESTER – I) END TERM EXAMINATION, OCTOBER - 2025****Statistical Analysis for Managers**

[Time: 1:00 Hours]

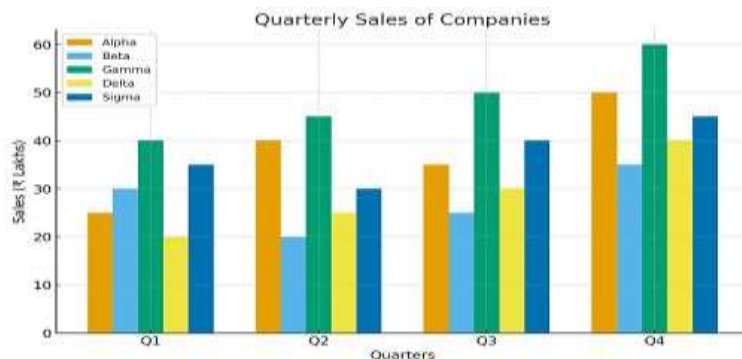
[Maximum Marks: 30]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
 (ii) Attempt questions from each section as per the directions.
 (iii) Make suitable assumptions wherever necessary.

SECTION-A**(Short Answer Type Questions, do not exceed 200 words)****Q.1. Attempt all parts of the following:****(3×5=15)**

- (a) A business researcher wants to conduct a survey to collect primary data from customers. What are the essential steps you would look for that guide the preparation of a questionnaire?
- (b) A research team in a large corporation wants to study employee satisfaction. One proposal suggests collecting feedback from all 10,000 employees, while another recommends surveying only 500 randomly chosen employees. If you were asked to identify the two approaches to data collection described by which terms would you recall, and how do they differ?
- (c) An HR department is curious about the reasons employees leave their jobs in the company. At the same time, it needs to compare its attrition rate with overall industry turnover statistics published in government labor reports. What different methods of collecting information would be suitable in these two cases?
- (d) The following table shows the quarterly sales (in ₹ lakhs) of five companies during a year:



- i) Suppose you are tasked with evaluating which company shows greater or lesser variation across quarters. Which statistical method would you select to make this comparison?
- ii) After applying this method, how would you determine which company is the most stable performer?
- (e) Suppose you are handed four different datasets: (i) blood groups of patients, (ii) ranks in a singing competition, (iii) temperatures in Fahrenheit, (iv) weights of school children. which measurement scales would you assign to each dataset and why?

SECTION-B**(Long Answer Type Questions)****Q.2. Attempt all questions of the following****(5×2=10)**

Expenditure (₹)	0–1000	1000–2000	2000–3000	3000–4000	4000–5000	5000–6000
Families	10	20	35	25	18	12

- (a) The government wants to know what value best represents the typical monthly spending. A welfare officer wants to know the expenditure level dividing families into two equal groups. Also Identify the top 25% of the expenditure threshold.

OR

- (b) As a officer you are required to segment families into four equal groups based on their monthly spending. To identify economically vulnerable households, also determine the expenditure thresholds for the bottom 10% and bottom 25% of families. These cut-offs help guide targeted financial support and policy planning.

Q.3.

Vehicles Owned	0	1	2	3	4	5
Households - Area A	6	12	15	9	5	3

Vehicles Owned	0	1	2	3	4	5
Households - Area B	4	10	12	7	5	2

- (a) A transport analyst wants to compare the stability of vehicle ownership between two towns (Dataset A and Dataset B). Explain which town shows greater consistency in vehicle ownership? Use the appropriate relative measure to justify your answer.

OR

Product	Chips	Cold Drinks	Chocolates	Biscuits	Soaps	Noodles	Juices
No. of Customers	2	5	12	18	14	6	3

- (b) As the marketing manager, you want to identify the product that was purchased in the largest concentration by customers during the promotional week. This will help in deciding which product should be highlighted in future combo packs and promotions. Based on the data, determine that product.

SECTION-C

Q.4. Attempt all parts of the following:

(1×5=5)

- 4(a) A marketing research firm was hired by a luxury retail chain to study customer behavior. The company wanted to know how Annual Income and Family Size affect a household's monthly spending on luxury products. Data was collected from 5 households, and the following regression output was generated:

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.985437
R Square	0.971086
Adjusted R Square	0.942171
Standard Error	0.671614
Observations	5

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	30.29787	15.14893	33.58475	0.02891448
Residual	2	0.902132	0.451066		
Total	4	31.2			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	6.158915	2.006583	3.069355	0.091769	-2.47471472	14.7925442	-2.4747147	14.79254418
Annual Income	0.429264	0.086199	4.979886	0.038038	0.058377172	0.80014996	0.05837717	0.80014996
Family size	-1.19671	0.274981	-4.35195	0.048955	-2.37985456	-0.0135563	-2.3798546	-0.01355628

- Q: As the marketing manager, you are considering targeting high-income households for luxury product promotions. Based on the regression output, explain whether income is a reliable predictor of luxury spending. The management team believes that larger families spend more on luxury goods. Does the regression analysis support or contradict this assumption? Provide justification. Estimate the monthly luxury spending of a household with an annual income of ₹20 lakhs and a family size of 4, using the regression results.

OR

4(b) Year	Interest Rate (%)	Investment (₹ Crores)
1	5	200
2	6	190
3	7	175
4	8	160
5	9	150
6	10	135
7	11	120
8	12	105

- Q: As the finance manager, you are concerned about how rising interest rates affect corporate investment decisions. Using the data, calculate the relationship between interest rates and investment, and explain whether the firm should accelerate investments at lower rates or defer them to later periods.